YZV416E Computer Vision

Istanbul Technical University

Department of AI and Data Engineering

Final Project Proposal Submission - Spring 2026 Term

Adapting Vision Models to Domain-Specific Tasks using Computer Vision Techniques

1. Topic Selection

Mark	Topic	Model	Dataset
[X]	Flow Estimation Robustness – RAFT vs GMFlow	RAFT & GMFlow	MPI-Sintel or Middlebury

Selected Topic: E - Flow Estimation Robustness – RAFT vs GMFlow

2. Team Information

- **Team name: LetItFlow**
- **Emin Acar** (major / role): Artificial Intelligence and Data Engineering / model pipeline,data curation
- **Batuhan Berk Balcı** (major / role): Artificial Intelligence and Data Engineering / model pipeline, evaluation
- **Hasan Yalçın Arıkanoğlu** (major / role): Artificial Intelligence and Data Engineering /model pipeline,report writing

3. Project Theme and Motivation

- **Problem:** We tackle dense optical flow estimation, predicting a per-pixel 2D motion vector (u,v) that maps every pixel in one video frame to its corresponding location in the next frame. Optical flow is a long-standing core Computer Vision problem with foundations going back to Horn and Schunck (1981), and it remains a key component of modern video understanding, used in tracking, action recognition, video stabilization, and novel-view synthesis. Concretely, our project is the Flow Estimation Robustness study (Topic E): we take two published, frozen state-of-the-art networks, RAFT (Teed & Deng, 2020) and GMFlow (Xu et al., 2022), and evaluate them in a zero-shot setting on the MPI-Sintel dataset (Butler et al., 2012) and the Middlebury optical flow benchmark (Baker et al., 2011), with particular focus on their performance under the challenging conditions described in Section 7.
- **Why it is challenging:** Optical flow is classically ill-posed: the brightness-constancy equation gives one constraint per pixel for two unknowns, so the flow is locally underdetermined, the well-known aperture problem, and real footage routinely violates brightness constancy through illumination change, motion blur, and occlusions (Horn & Schunck, 1981; Brox et al., 2004). Modern deep models resolve this ambiguity in very different ways, which is exactly what makes RAFT vs. GMFlow a revealing comparison: RAFT builds a 4D all-pairs correlation volume and iteratively refines the flow with a ConvGRU (Teed & Deng, 2020), while GMFlow replaces iterative refinement with a Transformer cross-attention pass that treats flow as global feature matching (Xu et al., 2022).
- **Expected learning and failure modes:** We expect both models to struggle in similar places. Weather, noise, and blur on the Sintel Final pass should hurt accuracy more than Clean. Illumination changes, shadows, and specularities break the brightness-match assumption both models rely on.

Small and fast-moving objects are easy to lose at low resolution, and sharp motion boundaries tend to get smoothed out. Textureless regions like sky, walls, and ground give both models very little to match. Our goal is to see which of these failure modes hits RAFT harder and which hits GMFlow harder.

4. Data Collection Plan

Describe:

- **Dataset source:** We use both the MPI-Sintel Optical Flow Dataset and the Middlebury Optical Flow benchmark. Sintel is the main benchmark reported by both RAFT and GMFlow, enabling comparison with prior work, while Middlebury provides a small real-world cross-dataset test.
- **Dataset size:** Sintel has 23 training clips (about 1,000 frames) with ground-truth flow and 12 test clips with hidden ground truth, all at 1024 × 436 resolution. Each frame comes in Clean, Final, and Albedo versions; we use Clean and Final, which align pixel-for-pixel and therefore support direct robustness comparison between RAFT and GMFlow. Middlebury is much smaller, containing only a limited number of annotated frame pairs, but it provides high-resolution real-world scenes often around 640 × 480 or higher depending on the sequence which makes it a useful complementary benchmark for cross-dataset evaluation.
- **Access and licensing:** Both datasets are free to download. Sintel is under CC BY 3.0 (free for academic and commercial use with citation) and Middlebury is under CC BY-NC (fine for a course project). Test-set ground truth is not public; you evaluate by uploading results to the official server. No special permissions or extra annotation are needed.
- **Preprocessing:** Each frame pair is loaded as an RGB tensor and normalized according to the respective model's input preprocessing. To accommodate architectural constraints, inputs are padded before inference and the predicted flow is cropped back to the original resolution.
- **Supplementary data:** We plan to use RobustSpring dataset to further test the robustness of the models since the dataset is constructed by applying 20 different corruptions to the test split of Spring dataset.

5. Example Model Input / Output Items

Image / Input ID	Input Description / Question	Expected Output / Answer	Concept / Facet Tested
sintel_clean_market_2_0015–16	Two consecutive frames (frame_0015.png, frame_0016.png) from MPI-Sintel Clean pass, sequence "market_2", a crowded scene with multi-agent motion and panning camera, 1024 × 436.	Dense per-pixel 2D flow field (u, v) at the input resolution.	In-distribution dense flow accuracy on Sintel Clean (baseline EPE for RAFT and GMFlow).
sintel_final_ambush_5_0020–21	The corresponding frame pair from the MPI-Sintel Final pass, with the same underlying scene motion but additional motion blur, defocus blur, and atmospheric effects.	Dense flow field; robustness is measured by the EPE increase relative to the corresponding Clean pair.	Robustness to rendering degradations in Clean-vs-Final evaluation.
middlebury_RubberWhale_frame10–11	Two Middlebury frames from *RubberWhale*, a real-captured sequence with	Dense flow evaluated zero-shot against ground truth, with emphasis on	Cross-dataset generalization to classic real-scene, small-

	hidden-texture ground truth.	non-occluded small-motion accuracy.	displacement optical flow.

6. Selected Model and Rationale

Indicate which model from the topic list you intend to use (or a justified alternative) and explain your choice.

- **Repository / checkpoint:** RAFT, https://github.com/princeton-vl/RAFT, checkpoint raft-sintel.pth (pretrained on Chairs → Things → Sintel). GMFlow, https://github.com/haofeixu/gmflow, checkpoint (pretrained on Chairs → Things → Sintel).
- **Key capabilities:** RAFT uses a 4D all-pairs correlation volume and an iterative ConvGRU refinement, which is known for high sub-pixel precision and sharp motion boundaries. GMFlow reformulates optical flow as a single-pass global matching problem using Transformer cross-attention, which handles large displacements more robustly. Together they represent the two dominant paradigms, iterative local refinement vs global matching, making them the natural pair for a robustness comparison.
- **Compute resources:** Google Colab. Since we are running inference only, both models fit on Colab without issues.
- **Evaluation strategy:** Zero-shot evaluation on the Sintel training set, using each model's official pretrained checkpoint with no fine-tuning.

7. Expected Challenges and Risks

Briefly note potential issues and how you plan to address or mitigate them:

- Weather, illumination, and sensor corruptions: flow models lose accuracy under noise, blur, rain, and fog. Mitigation: we measure the Clean-vs-Final EPE gap on Sintel, which already contains motion blur, defocus, and atmosphere.
- Illumination changes, shadows, and specularities: when pixel brightness changes between frames, appearance matching breaks down. Mitigation: Sintel Final already includes these effects, so the issue shows up in our evaluation.
- Small and fast-moving objects: tiny fast-moving objects are easy to lose in downsampling or confuse with the background. Mitigation: we report per-sequence EPE on fast-motion clips (e.g. market_2, ambush_5) instead of only averages.
- Corners and sharp motion boundaries: models tend to smooth flow across moving-object edges and blur the boundary. Mitigation: we report non-occluded EPE and visually check flow near object silhouettes.
- Textureless regions: on flat surfaces like sky, walls, or ground, there is no texture to match and flow becomes ambiguous. Mitigation: we flag these regions in our qualitative analysis as a known limitation.

8. Preliminary Timeline

Provide a short week-by-week outline of what you will accomplish up to June.

Since this project is inference-only (no fine-tuning), our timeline focuses on setup, running the two models on Sintel, comparing them, and writing up results.

Phase / Weeks	Tasks	Deliverable
Weeks 10-11	Set up Colab environment; clone RAFT and GMFlow repos; download pretrained checkpoints and Sintel dataset; run a first forward pass on one frame pair	Both models running with a sample flow output

Weeks 11–12	Write loading + preprocessing pipeline (per-model normalization, reflection padding); run both models on Sintel Clean	Clean-pass EPE numbers for RAFT and GMFlow
Weeks 12–13	Run both models on Sintel Final; compute Clean-vs-Final EPE delta; run Middlebury zero-shot check	Robustness comparison table
Weeks 13-14	Per-sequence error analysis; flow-field visualizations; qualitative failure cases	Plots and error analysis
Weeks 14–15	Write report; clean up code; prepare final presentation	Final report + code repository

9. References

Cite papers, datasets, and repositories you have already reviewed. Some are listed below for reference. Add topic-specific references:

- Teed, Z. & Deng, J. (2020). RAFT: Recurrent All-Pairs Field Transforms for Optical Flow. ECCV 2020. https://github.com/princeton-vl/RAFT

- Xu, H. et al. (2022). GMFlow: Learning Optical Flow via Global Matching. CVPR 2022. https://github.com/haofeixu/gmflow

- Butler, D. J., Wulff, J., Stanley, G. B., & Black, M. J. (2012). A Naturalistic Open Source Movie for Optical Flow Evaluation. ECCV 2012. http://sintel.is.tue.mpg.de

- Baker, S., Scharstein, D., Lewis, J. P., Roth, S., Black, M. J., & Szeliski, R. (2011). A Database and Evaluation Methodology for Optical Flow. International Journal of Computer Vision, 92(1).

- Horn, B. K. P., & Schunck, B. G. (1981). Determining Optical Flow. Artificial Intelligence, 17(1–3).

- Brox, T., Bruhn, A., Papenberg, N., & Weickert, J. (2004). High Accuracy Optical Flow Estimation Based on a Theory for Warping. In Computer Vision – ECCV 2004.

- Oei, V., Schmalfuss, J., Mehl, L., Bartsch, M., Agnihotri, S., Keuper, M., Bulling, A., & Bruhn, A. (2026). RobustSpring: Benchmarking Robustness to Image Corruptions for Optical Flow, Scene Flow and Stereo. In International Conference on Learning Representations (ICLR).

- Mehl, L., Schmalfuss, J., Jahedi, A., Nalivayko, Y., & Bruhn, A. (2023). Spring: A High-Resolution High-Detail Dataset and Benchmark for Scene Flow, Optical Flow and Stereo. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR).